

Self-Supervised Representation Learning for Early Warning and Risk Assessment of Cascading Failures in Complex Energy Networks

Author: Minglei Zhou

Updated: Mar 8, 2026

1 Research Background

Modern power systems are evolving into complex energy networks characterized by strong topological coupling, increasing renewable penetration, and significant operational uncertainty. Under such conditions, local disturbances may propagate through network topology and power flow redistribution, eventually leading to cascading failures and large-scale system instability.

Recent studies have increasingly applied machine learning techniques to cascading failure analysis, including vulnerability identification, failure propagation prediction, and blackout risk estimation. However, most existing approaches rely heavily on **supervised learning**, which requires large amounts of labeled cascading failure data. In practice, such labels are difficult and expensive to obtain because cascading failures are rare events and often require detailed simulation or historical analysis.

At the same time, several large-scale public datasets have recently become available for power system research. These datasets contain large volumes of operational samples that can potentially be used for **representation learning**, even though they may not directly provide cascading failure labels.

This motivates the central research question of this study:

Can self-supervised representation learning extract transferable graph representations from large-scale unlabeled operational data and improve early warning and risk assessment for cascading failures in complex energy networks?

2 Research Gap

Existing research reveals three key limitations.

2.1 Heavy reliance on labeled cascading failure data

Most machine learning approaches formulate cascading failure prediction directly as supervised classification or regression problems. However, cascading failure labels are scarce, highly imbalanced, and costly to generate.

As a result, many models struggle with **limited training data** and **poor generalization**.

2.2 Lack of representation learning from operational data

Modern power systems generate large amounts of operational data such as power flows, voltages, and loading conditions. However, most studies do not fully exploit these large-scale unlabeled datasets to learn general system representations.

Instead, they train models directly on small labeled datasets.

2.3 Limited robustness under topology perturbations

Power system topology and operating conditions frequently change due to line outages, maintenance actions, or renewable fluctuations. Many existing models show limited robustness under such perturbations.

Therefore, learning **topology-aware and transferable representations** remains an open challenge.

3 Research Objectives

The main objective of this research is to develop a **self-supervised graph representation learning framework** for cascading failure analysis in complex energy networks.

The framework aims to learn transferable graph-temporal representations from large-scale operational data and apply them to two downstream tasks:

- cascading failure **early warning**
- cascading failure **risk assessment**

More specifically, this research aims to:

1. construct a unified graph-based representation of complex energy networks;
2. learn transferable representations using self-supervised graph pretraining;
3. transfer the learned representations to temporal prediction models for cascading failure analysis;
4. evaluate the framework under topology perturbations and varying operating conditions.

4 Data Sources and Their Roles

This research integrates three publicly available datasets that serve complementary roles in the proposed framework.

Dataset	Role	Description
PGLib-OPF	Network templates	Standardized AC optimal power flow test cases
OPFData	Pretraining dataset	Large-scale AC-OPF operational samples
PowerGraph	Downstream benchmark	Graph-learning tasks including cascading failure prediction

4.1 OPFData for Self-Supervised Pretraining

OPFData provides approximately **300,000 samples per power grid case**, covering multiple networks ranging from **14-bus systems to large-scale networks exceeding 13,000 buses**. In total, the dataset contains **millions of AC-OPF samples** generated under varying operating conditions and topology perturbations.

These samples include:

- node-level electrical quantities (voltage magnitude, voltage angle)
- generator outputs
- branch power flows
- line loading conditions

Although OPFData does not directly provide cascading failure labels, it offers a large volume of operational states that can be used for **self-supervised representation learning**.

In this research, OPFData will be used primarily for **graph representation pretraining**.

4.2 PowerGraph for Downstream Tasks

While OPFData provides large-scale operational samples, cascading failure prediction requires labeled downstream tasks. The **PowerGraph benchmark** provides graph learning tasks specifically designed for power system applications, including cascading failure analysis.

In this research, PowerGraph will be used for:

- early warning classification

- cascading failure risk estimation

Thus, the datasets play complementary roles:

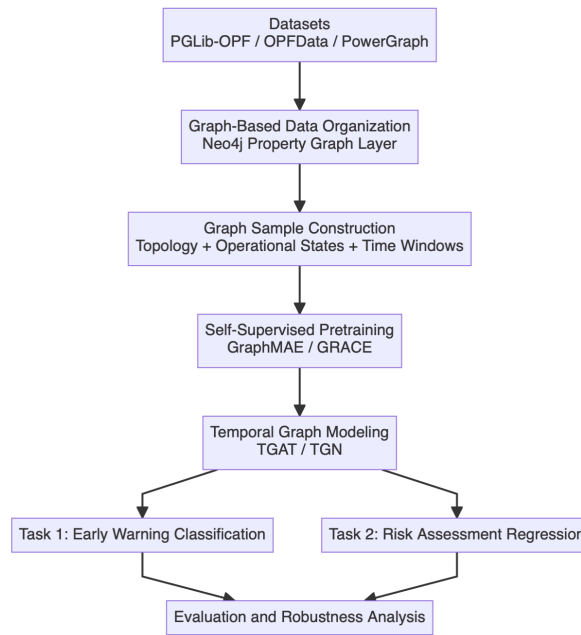
OPFData → representation learning

PowerGraph → downstream prediction tasks

5 Methods

The proposed framework consists of four major stages.

5.1 Technical Route Diagram



5.2 Graph-Based Data Organization

Complex energy networks consist of multiple types of entities, including buses, transmission lines, generators, and loads. These components form a highly relational structure that is naturally represented as a graph.

In this study, a **property graph model** implemented using **Neo4j** is used as a **data organization layer**.

It serves three main purposes:

1. integrating heterogeneous datasets into a unified graph representation;
2. supporting graph sample construction and temporal subgraph extraction;
3. providing standardized inputs for machine learning models.

It is important to note that Neo4j is **not used as a predictive model**. Instead, it functions solely as a data management infrastructure.

5.3 Self-Supervised Representation Learning

The core methodological component of this research is **graph self-supervised representation learning**.

Instead of relying solely on labeled cascading failure data, the model first learns general graph representations from large-scale operational samples.

Two representative methods are considered.

Model	Type	Mechanism
GraphMAE	Generative self-supervision	Mask node features and reconstruct them
GRACE	Contrastive self-supervision	Learn invariant representations across graph augmentations

GraphMAE is chosen as the primary pretraining method because power system graphs contain rich continuous attributes such as voltage levels, power flows, and line loading. Feature reconstruction objectives are therefore well aligned with the characteristics of power system data.

GRACE is included as a baseline to compare generative and contrastive self-supervised learning strategies.

5.4 Temporal Graph Modeling

Cascading failures are dynamic processes involving disturbance occurrence and sequential propagation. Static graph representations alone cannot fully capture these dynamics.

Therefore, the pretrained representations will be transferred to **temporal graph neural networks**.

Two candidate models are considered.

Model	Mechanism
TGAT	Time encoding with attention
TGN	Memory-based dynamic graph learning

These models enable the framework to capture temporal dependencies and event-driven system dynamics.

5.5 Downstream Tasks

Two downstream tasks are considered.

5.5.1 Early Warning

A classification task that predicts the system risk level before cascading failures occur.

Possible outputs:

- safe state
- pre-cascade state
- high-risk state

5.5.2 Risk Assessment

A regression task that estimates the severity or risk level of cascading failures.

6 Evaluation Strategy

The framework will be evaluated from three perspectives.

6.1 Predictive Performance

Classification metrics:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

Regression metrics:

- MAE
- RMSE
- MSE
- (R^2)

6.2 Representation Learning Effectiveness

Three configurations will be compared:

1. downstream model without pretraining
2. GRACE + downstream model
3. GraphMAE + downstream model

This comparison evaluates whether self-supervised pretraining improves prediction performance.

6.3 Robustness and Generalization

Additional experiments will test the model under:

- topology perturbations
- varying operating conditions
- renewable generation fluctuations

These scenarios will primarily use OPFData perturbation samples.

7 Expected Contributions

This research is expected to contribute in three aspects.

First, it proposes a **self-supervised representation learning framework** for cascading failure analysis.

Second, it explores how large-scale unlabeled operational data can improve cascading failure prediction.

Third, it evaluates the robustness of learned representations under topology perturbations and dynamic operating conditions.

8 Conclusion

This research proposes a unified framework that integrates graph-based data organization, self-supervised representation learning, and temporal graph modeling for cascading failure analysis in complex energy networks.

By leveraging large-scale operational data and transferring learned representations to downstream prediction tasks, the proposed framework aims to improve early warning and risk assessment under conditions of label scarcity and dynamic system behavior.